

Avi Vadali

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EDUCATION

- **Massachusetts Institute of Technology** September 2026 - Present
Ph.D. in Physics, Condensed Matter Theory Cambridge, MA, USA
- **California Institute of Technology** September 2022 – June 2026
B.S. in Physics; GPA 4.2/4.3 Pasadena, CA, USA
 - Thesis: *Statistical mechanics and symmetries of anyon-nucleating deformations* (advised by Jason Alicea).

PUBLICATIONS

- [1] **Vadali, A.**, Soejima, T. (2026). *Berry curvature from plane-wave mixing: a three-band model with emergent symmetries*. (In preparation).
- [2] **Vadali, A.**, Vanhove, R., Verresen, R., Alicea, J., Sala, P. (2026). *Statistical Mechanics and Symmetries of Non-Abelian Anyon Proliferation: From Deformation to Decoherence*. arXiv preprint arXiv:2606.12527.
- [3] **Vadali, A.**, Turner, A. (2026). *Equality of cycle lengths in one- and two-dimensional σ automata*. Theoretical Computer Science, 1074, 115897.
- [4] **Vadali, A.**, Wang, Z., Dua, A., Shirley, W., Chen, X. (2024). *Composite subsystem symmetries and decoration of sub-dimensional excitations*. SciPost Phys., 17, 071.
- [5] **Vadali, A.**, Kshirsagar, R., Shyamsundar, P., et al. (2024). *Quantum circuit fidelity estimation using machine learning*. Quantum Mach. Intell., 6, 1.

RESEARCH EXPERIENCE

- **Jason Alicea, California Institute of Technology**  October 2024 – June 2026
Undergraduate Researcher Pasadena, CA, USA
 - Identified and characterized phase transitions in D_4 non-Abelian topological order using Monte Carlo simulations; analyzed near-critical behavior using effective quantum field theory descriptions. Published as [2].
 - Contributed talk at APS March Meeting 2026; presented at Caltech SURF Seminar Day 2025.
- **Ashvin Vishwanath, Harvard University**  June 2024 – May 2026
Undergraduate Researcher Cambridge, MA, USA
 - Developed a simplified model of a spinor in a moiré potential that captures emergent symmetries and quadratic band touchings at topological critical points, enabling mean-field study of topological crystals. Manuscript in preparation [1].
 - Presented at Caltech SURF Seminar Day 2024.
- **Xie Chen, California Institute of Technology**  April 2022 – December 2023
Undergraduate Researcher Pasadena, CA, USA
 - Developed a “remote detectability” method to deduce the bound flux of combined $\mathbb{Z}_2$ gauge theories without performing the full gauging procedure. Published as [4].
 - Presented at Caltech SURF Seminar Day 2023.
- **Gabriel Perdue, Fermi National Accelerator Laboratory**  October 2021 – December 2022
High School Researcher (remote) Batavia, IL, USA
 - Used machine learning to predict fidelities of quantum circuits under custom noise models; trained on Clifford circuits and extrapolated to non-Clifford circuits. Published as [5].
- **Ari Turner, Technion – Israel Institute of Technology**  November 2020 – February 2025
High School / Undergraduate Researcher (remote) Haifa, Israel
 - Proved equality between cycle lengths of a class of 1D and 2D σ -automata and bounded periods of higher-dimensional automata; derived a closed-form Jordan structure for the Kronecker sum of Jordan blocks over $\text{GF}(2)$. Published as [3].

TALKS & PRESENTATIONS

- [1] *Instability of topological order to nucleation of multiple non-Abelian anyon species*. Contributed talk, APS March Meeting (2026).
- [2] *Instability of topological order to nucleation of multiple non-Abelian anyon species*. Contributed talk, Caltech SURF Seminar Day (2025).
- [3] *Berry curvature from plane-wave mixing: a three-band model with emergent symmetries*. Contributed talk, Caltech SURF Seminar Day (2024).
- [4] *Composite subsystem symmetries and decoration of sub-dimensional excitations*. Contributed talk, Caltech SURF Seminar Day (2023).

FELLOWSHIPS & AWARDS

Graduate Fellowships & Awards

C.M. Clay and Henry W. Kendall (First Year) Fellowships, MIT	August 2026
NSF Graduate Research Fellowship (GRFP)	April 2026
Paul & Daisy Soros Fellowship	March 2026
Chair's Fellowship, CU Boulder (<i>declined</i>)	March 2026
Stanford Graduate Fellowship, Stanford (<i>declined</i>)	February 2026
Yoichiro Nambu Fellowship, U. Chicago (<i>declined</i>)	January 2026

Undergraduate Fellowships & Awards

Richard P. Feynman Prize in Theoretical Physics, Caltech	June 2026
Doris S. Perpall Speaking Competition, Finalist, Caltech	November 2025
Haren Lee Fisher Memorial Award in Junior Physics, Caltech	June 2025
Bill Davis Summer Undergraduate Research Fellowship (Caltech)	May 2025
Caltech Summer Undergraduate Research Fellowship (at Harvard)	May 2024
Barry Goldwater Fellowship Nominee, Caltech	January 2024
Ernest R. Roberts Summer Undergraduate Research Fellowship (Caltech)	May 2023

TEACHING

- **California Institute of Technology** April 2024 – Present
Undergraduate Teaching Assistant Pasadena, CA, USA
 - Ph 22, *Computational Physics Laboratory* (Spring 2025): led recitations, graded assignments.
 - Ph 6, *Physics Laboratory* (Winter 2025): led recitations, graded lab reports, and supported students with experimental apparatuses.
 - Ph 2a, *Waves and Oscillations* (Fall 2024): graded assignments and exams.
 - Ph 7, *Physics Laboratory* (Spring 2024): graded prelabs and supported students with experimental apparatuses.

SERVICE & OUTREACH

- **Caltech Undergraduate Physics Club**  September 2023 – Present
Vice President Pasadena, CA, USA
 - Engage with club members and address feedback to shape programming.
 - Co-organize physics colloquia and propose faculty speakers based on member interest.

SKILLS

- **Programming & Tools:** Python, Julia, Mathematica, GAP.